XML XSD Structuur

CRM

Architectuur

[registratie / andere systemen]

↓ XML

crm.incoming

[kassa]

kassa.payments → CRM (receiver.js)

kassa.incoming (sender.js)

crm.to.facturatie (sender.js)

crm.to.mailing (sender.js)

crm.dead-letter (receiver.js)

Salesforce (sfConnection.js)

Supabase (supabaseClient.js)

Algemene berichtstructuur

XML :

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>...</message_id>
    <version>2.0</version>
    <type>...</type>
    <timestamp>2026-04-01T13:44:00.000Z</timestamp>
    <source>...</source>
    <!-- optioneel -->
    <correlation_id>...</correlation_id>
  </header>
```

```
  <body>
    <!-- berichttype-specifieke inhoud -->
  </body>
</message>
```

Algemene XSD Header

XSD

Algemene XSD Header (v2.0) Elk bericht in dit document volgt de standaard header-validatie: - **version**: `Altijd "2.0"` . **message_id**: `UUID string.` **timestamp**: `ISO 8601 (UTC)` . **source**: `Systeemnaam (lowercase)` . En dan vul je bij de acties zelf alleen nog de Body-regels in: **Voorbeeld** bij `session_update` : XSD: Specifieke body-regels: **start_time** en **end_time** moeten **geldige ISO-timestamps zijn**. **status** is **verplicht**.

Naar CRM (Inkomend)

Frontend → CRM

new_registration

Registratie in salesforce (info van frontend)

XML

```
<message>
  <header>
    <message_id>test-abc-123</message_id>
    <version>2.0</version>
    <type>new_registration</type>
    <timestamp>2026-04-01T10:00:00.000Z</timestamp>
    <source>registratie</source>
  </header>
  <body>
    <customer>
      <email>jan@example.com</email>
      <first_name>Jan</first_name>
      <last_name>Peeters</last_name>
      <date_of_birth>1995-06-15</date_of_birth>
      <type>private</type>                    <!-- "private" of "company" -->
      <user_id>user-001</user_id>
      <badge_id>BADGE-001</badge_id>          <!-- optioneel -->
      <is_company_linked>false</is_company_linked> <!-- optioneel -->
      <company_name>Acme BV</company_name>    <!-- optioneel -->
```

```
      <vat_number>BE0123456789</vat_number>       <!-- optioneel -->
      <registration_date>2026-04-01</registration_date> <!-- optioneel -->
      <address>                                   <!-- optioneel -->
        <street>Kerkstraat</street>
        <number>12</number>
        <postal_code>2000</postal_code>
        <city>Antwerpen</city>
        <country>BE</country>
      </address>
      <registration_fee>                          <!-- optioneel -->
        <amount currency="eur">150.00</amount>
        <paid>false</paid>
      </registration_fee>
    </customer>
    <session_id>sess-42</session_id>              <!-- optioneel -->
    <payment_due>                                 <!-- optioneel -->
      <amount>150.00</amount>
      <status>unpaid</status>
    </payment_due>
  </body>
</message>
```

XSD

`user_id`, `email`, `first_name`, `last_name`, `date_of_birth`. De `date_of_birth` moet in `YYYY-MM-DD` formaat zijn voor de leeftijdcalculator.

Facturatie → CRM

invoice_status

maakt task in salesforce

XML

```
<message>
  <header><type>invoice_status</type>...</header>
  <body>
    <email>jan@example.com</email>   <!-- optioneel -->
    <invoice>
      <id>INV-2026-001</id>
```

```
    <status>paid</status>
    <amount_paid>150.00</amount_paid>
  </invoice>
 </body>
</message>
```

XSD

Verplicht: `invoice.id` en `invoice.status`. Dit bericht koppelt de betalingsstatus uit het boekhoudpakket aan de klant in het CRM via het `email` veld (unieke identifier).

payment_registered

Facturatie → CRM

XML

```
<message>
 <header>
  <type>payment_registered</type>
  <source>facturatie</source>
  ...
 </header>
 <body>
  <user_id>user-001</user_id>          <!-- optioneel, OF email -->
  <email>jan@example.com</email>       <!-- optioneel -->
  <payment_context>registration</payment_context> <!-- "registration" of "consumption" -->
  <invoice>
   <id>INV-2026-001</id>
   <amount_paid currency="eur">150.00</amount_paid>
   <due_date>2026-04-15</due_date>
   <status>paid</status>
  </invoice>
  <transaction>
   <payment_method>bancontact</payment_method>
  </transaction>
 </body>
</message>
```

XSD

Verplicht: `invoice.id` en `invoice.status` (waarde: 'paid'). `payment_context` moet 'registration' of 'consumption' zijn.

IoT → CRM

badge_scanned

IoT → CRM

XML

```
<message>
  <header><type>badge_scanned</type><source>iot</source>...</header>
  <body>
    <badge_id>BADGE-XYZ-001</badge_id>
    <email>jan@example.com</email>   <!-- optioneel -->
    <scan_type>entry</scan_type>
    <location>Ingang A</location>
  </body>
</message>
```

XSD

Planning → CRM

session_update

maakt task in salesforce

XML

```
<message>
  <header><type>session_update</type>...</header>
  <body>
    <session_name>Keynote: AI 2026</session_name>
    <speaker>Prof. De Smet</speaker>
    <start_time>2026-04-01T09:00:00Z</start_time>
    <end_time>2026-04-01T10:00:00Z</end_time>
    <status>ongoing</status>
  </body>
</message>
```

XSD

Verplicht: `session_name` en `status`. De velden `start_time` en `end_time` moeten voldoen aan het ISO8601-formaat (`YYYY-MM-DDTHH:MM:SSZ`) om correcte tijdstippen in de Salesforce-taken te garanderen.

Kassa → CRM

consumption_order

Kassa → CRM

XML

```
<message>
  <header><type>consumption_order</type><source>kassa</source>...</header>
  <body>
    <is_anonymous>false</is_anonymous>
    <customer>
      <email>jan@example.com</email>
      <user_id>user-001</user_id>
    </customer>
    <items>
      <item>
        <description>Koffie</description>
        <quantity>2</quantity>
        <unit_price currency="eur">2.50</unit_price>
      </item>
    </items>
  </body>
</message>
```

XSD

Verplicht: `items` lijst met minimaal één `item`. Elk `item` moet een `description`, `quantity` (integer) en `unit_price` (decimal) hebben. `is_anonymous` is een boolean.

badge_assigned

Kassa → CRM

XML

```
<message>
  <header><type>badge_assigned</type>...</header>
  <body>
    <badge_id>BADGE-XYZ-001</badge_id>
    <user_id>user-001</user_id>
  </body>
</message>
```

XSD

Verplicht: `badge_id` en `scan_type` (waarde: 'entry' of 'exit'). Locatie is optioneel. `badge_id` en `user_id`. Dit is een kritieke koppeling: de `badge_id` wordt als string opgeslagen op het `Member__c` record in Salesforce om toegang bij IoT-scanners te verlenen.

refund_processed

Kassa → CRM

XML

```
<message>
  <header><type>refund_processed</type>...</header>
  <body>
    <user_id>user-001</user_id>
    <refund_type>partial</refund_type>
    <original_transaction_id>TX-999</original_transaction_id>
    <refund>
      <amount currency="eur">25.00</amount>
      <method>bancontact</method>
      <reason>duplicate</reason>
      <description>Dubbele betaling</description>
    </refund>
    <new_wallet_balance currency="eur">10.00</new_wallet_balance> <!-- optioneel --
>
  </body>
</message>
```

XSD

Verplicht: `user_id`, `refund_type` en `refund.amount`. De `original_transaction_id` is nodig voor de audit-trail in Salesforce. Bedragen moeten voorzien zijn van een `currency` attribuut.

invoice_request

Kassa → CRM

XML

```
<message>
  <header>
    <type>invoice_request</type>
    <correlation_id>corr-abc</correlation_id>
```

```
    ...
  </header>
  <body>
    <user_id>user-001</user_id>
    <invoice_data>
      <first_name>Jan</first_name>
      <last_name>Peeters</last_name>
      <email>jan@example.com</email>
      <vat_number>BE0123456789</vat_number>  <!-- optioneel -->
      <address>                              <!-- optioneel -->
        <street>Kerkstraat</street>
        <number>12</number>
        <postal_code>2000</postal_code>
        <city>Antwerpen</city>
      </address>
    </invoice_data>
  </body>
</message>
```

XSD

Mailing → CRM

mailing_status

Mailing → CRM

XML

```
<message>
  <header><type>mailing_status</type>...</header>
  <body>
    <mailing_id>MAIL-001</mailing_id>
    <status>delivered</status>
    <delivered>950</delivered>
    <bounced>12</bounced>
  </body>
</message>
```

XSD

Verplicht: `mailing_id` en `status`. De velden `delivered` en `bounced` moeten gehele getallen (integers) zijn. Wordt gebruikt voor het bijwerken van campagne-statistieken in Salesforce.

Van CRM (Uitgaand)

CRM → Kassa

new_registration

CRM → Kassa (kassa.incoming)

XML

```xml
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>reg-crm-{uuid}</message_id>
    <type>new_registration</type>
    <source>crm</source>
    <timestamp>2026-04-01T10:00:01.000Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <customer>
      <email>jan@example.com</email>
      <contact>
        <first_name>Jan</first_name>
        <last_name>Peeters</last_name>
      </contact>
      <company_name>Acme BV</company_name>   <!-- optioneel -->
      <type>private</type>
      <vat_number>BE0123456789</vat_number>  <!-- optioneel -->
      <user_id>user-001</user_id>
      <age>30</age>                          <!-- berekend uit date_of_birth! -->
    </customer>
    <payment_due>
      <amount>150.00</amount>
      <status>unpaid</status>   <!-- altijd "paid" of "unpaid" — zie hieronder -->
    </payment_due>
```

```
  </body>
</message>
```

XSD

Bevat berekend veld `age` (integer). De `payment_due.status` wordt door de sender genormaliseerd naar exact 'paid' of 'unpaid'.

profile_update

CRM → Kassa (kassa.incoming)

XML

```
<message>
  <header>
    <message_id>prof-crm-{uuid}</message_id>
    <type>profile_update</type>
    <source>crm</source>
    <timestamp>...</timestamp>
    <version>2.0</version>
    <correlation_id>corr-abc</correlation_id>  <!-- optioneel -->
  </header>
  <body>
    <user_id>user-001</user_id>
    <email>jan@example.com</email>
    <contact>
      <first_name>Jan</first_name>
      <last_name>Peeters</last_name>
    </contact>
    <company_name>Acme BV</company_name>       <!-- optioneel -->
    <age>30</age>
    <type>private</type>
    <vat_number>BE0123456789</vat_number>      <!-- optioneel -->
  </body>
</message>
```

XSD

Verplicht: `user_id`. Bevat de meest recente contactgegevens (`first_name`, `last_name`, `email`). Het veld `age` wordt door de CRM opnieuw berekend op basis van de geboortedatum in Salesforce voordat het naar de kassa gaat.

cancel_registration

CRM → KASSA (kassa.incoming)

XML

```
<message>
  <header>
    <message_id>cancel-crm-{uuid}</message_id>
    <type>cancel_registration</type>
    <source>crm</source>
    <timestamp>...</timestamp>
    <version>2.0</version>
    <correlation_id>...</correlation_id>  <!-- optioneel -->
  </header>
  <body>
    <user_id>user-001</user_id>
    <session_id>sess-42</session_id>
  </body>
</message>
```

XSD

Verplicht: `user_id` en `session_id`. Dit bericht dient als trigger voor de kassa om de toegangsrechten voor een specifieke sessie of badge onmiddellijk in te trekken.

CRM → Facturatie

invoice_request

CRM → Facturatie (crm.to.facturatie)

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>inv-crm-{uuid}</message_id>
    <version>2.0</version>
    <type>invoice_request</type>
    <timestamp>...</timestamp>
    <source>crm</source>
    <correlation_id>...</correlation_id>  <!-- optioneel -->
  </header>
```

```
  <body>
    <customer>
      <email>jan@example.com</email>
      <first_name>Jan</first_name>
      <last_name>Peeters</last_name>
      <phone>+32 478 000 000</phone>    <!-- optioneel -->
    </customer>
    <invoice>
      <description>Invoice request for user user-001</description>
      <amount currency="eur">0</amount>
      <due_date>2026-04-01</due_date>
      <invoice_number>INV-001</invoice_number>  <!-- optioneel -->
    </invoice>
    <items>
      <item>
        <description>...</description>
        <quantity>1</quantity>
        <unit_price currency="eur">...</unit_price>
        <vat_rate>21</vat_rate>
        <sku>...</sku>  <!-- optioneel -->
      </item>
    </items>
  </body>
</message>
```

XSD

Verplicht: `customer.email`. De `items` lijst bevat per item een `vat_rate` (integer, bijv. 21) en een `unit_price`.

CRM → Centrale Monitoring Tool

heartbeat

heartbeat.js

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
```

```
    <message_id>hb-crm-{uuid}</message_id>
    <version>2.0</version>
    <type>heartbeat</type>
    <timestamp>2026-04-01T10:00:00.000Z</timestamp>
    <source>crm</source>
  </header>
  <body>
    <status>online</status>   <!-- "online" | "degraded" | "offline" -->
    <service>crm</service>
    <uptime>3600</uptime>     <!-- seconden -->
  </body>
</message>
```

XSD

Verplicht: `status` (keuze uit: `online`, `degraded`, `offline`), `service` (waarde: 'crm') en `uptime` (integer in seconden). Dit schema wijkt af van de business-logica en dient enkel voor systeembeheer.

CRM → Mailing

mailing_status

CRM → Mailing (crm.to.mailing)

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>mail-crm-{uuid}</message_id>
    <version>2.0</version>
    <type>mailing_status</type>
    <timestamp>...</timestamp>
    <source>crm</source>
    <correlation_id>...</correlation_id>  <!-- optioneel -->
  </header>
  <body>
    <mailing>
      <subject>Welkom bij het event!</subject>
      <template_id>TPL-001</template_id>
```

```
    <from_address>info@event.be</from_address>  <!-- optioneel -->
    <reply_to>noreply@event.be</reply_to>       <!-- optioneel -->
  </mailing>
  <recipients>
    <recipient>
      <email>jan@example.com</email>
      <first_name>Jan</first_name>
      <last_name>Peeters</last_name>
      <language>NL</language>  <!-- optioneel -->
    </recipient>
  </recipients>
 </body>
</message>
```

XSD

Verplicht: `mailing.template_id` en de lijst `recipients`. Elk `recipient` object moet minimaal een geldig `email` adres bevatten.

Facturatie

Naar Facturatie (Inkomend)

CRM => Facturatie

consumption_order

Bevat de details van een nieuwe bestelling of verbruik die gefactureerd moet worden in FossBilling.

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>f47ac10b-58cc-4372-a567-0e02b2c3d479</message_id>
    <version>2.0</version>
    <type>consumption_order</type>
    <timestamp>2026-03-29T18:30:00Z</timestamp>
    <source>crm_system</source>
  </header>
  <body>
    <customer>
```

```
      <id>12345</id>
      <is_company_linked>true</is_company_linked>
      <company_id>FOSS-CUST-102</company_id>
      <company_name>Bedrijf BV</company_name>
      <email>info@bedrijf.be</email>
      <address>
        <street>Kiekenmarkt</street>
        <number>42</number>
        <postal_code>1000</postal_code>
        <city>Brussel</city>
        <country>be</country>
      </address>
    </customer>
    <payment_method>online</payment_method>
    <items>
      <item>
        <id>BEV-001</id>
        <description>Koffie</description>
        <quantity>2</quantity>
        <unit_price currency="eur">2.50</unit_price>
        <vat_rate>21</vat_rate>
      </item>
    </items>
  </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" elementFormDefault="qualified">
  <!-- ROOT -->
  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
```

```
      <xs:element name="body" type="ConsumptionOrderBodyType"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>
<!-- HEADER -->
<xs:complexType name="HeaderType">
  <xs:sequence>
    <xs:element name="message_id" type="UuidV4"/>
    <xs:element name="version" type="xs:string" fixed="2.0"/>
    <xs:element name="type" type="xs:string" fixed="consumption_order"/>
    <xs:element name="timestamp" type="xs:dateTime"/>
    <xs:element name="source" type="xs:string"/>
    <xs:element name="correlation_id" type="UuidV4" minOccurs="0"/>
  </xs:sequence>
</xs:complexType>
<!-- BODY -->
<xs:complexType name="ConsumptionOrderBodyType">
  <xs:sequence>
    <xs:element name="customer" type="CustomerType"/>
    <xs:element name="payment_method" type="PaymentMethodType"/>
    <xs:element name="items" type="ItemsType"/>
  </xs:sequence>
</xs:complexType>
<!-- CUSTOMER -->
<xs:complexType name="CustomerType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="is_company_linked" type="xs:boolean"/>
    <xs:element name="company_id" type="xs:string" minOccurs="0"/>
    <xs:element name="company_name" type="xs:string" minOccurs="0"/>
    <xs:element name="email" type="xs:string"/>
    <xs:element name="address" type="AddressType"/>
  </xs:sequence>
</xs:complexType>
```

```
<!-- ADDRESS -->
<xs:complexType name="AddressType">
  <xs:sequence>
    <xs:element name="street" type="xs:string"/>
    <xs:element name="number" type="xs:string"/>
    <xs:element name="postal_code" type="xs:string"/>
    <xs:element name="city" type="xs:string"/>
    <xs:element name="country" type="CountryCodeType"/>
  </xs:sequence>
</xs:complexType>
<!-- ITEMS -->
<xs:complexType name="ItemsType">
  <xs:sequence>
    <xs:element name="item" type="ItemType" maxOccurs="unbounded"/>
  </xs:sequence>
</xs:complexType>
<!-- ITEM -->
<xs:complexType name="ItemType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="description" type="xs:string"/>
    <xs:element name="quantity" type="xs:positiveInteger"/>
    <xs:element name="unit_price" type="AmountType"/>
    <xs:element name="vat_rate" type="VatRateType"/>
  </xs:sequence>
</xs:complexType>
<!-- AMOUNT -->
<xs:complexType name="AmountType">
  <xs:simpleContent>
    <xs:extension base="xs:decimal">
      <xs:attribute name="currency" type="CurrencyCodeType" use="required"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>
```

```
<!-- SIMPLE TYPES -->
<xs:simpleType name="UuidV4">
  <xs:restriction base="xs:string">
    <xs:pattern value="[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}"/>
  </xs:restriction>
</xs:simpleType>
<xs:simpleType name="CountryCodeType">
  <xs:restriction base="xs:string">
    <xs:pattern value="[a-z]{2}"/>
  </xs:restriction>
</xs:simpleType>
<xs:simpleType name="CurrencyCodeType">
  <xs:restriction base="xs:string">
    <xs:pattern value="[a-z]{3}"/>
  </xs:restriction>
</xs:simpleType>
<xs:simpleType name="VatRateType">
  <xs:restriction base="xs:integer">
    <xs:enumeration value="6"/>
    <xs:enumeration value="12"/>
    <xs:enumeration value="21"/>
  </xs:restriction>
</xs:simpleType>
<xs:simpleType name="PaymentMethodType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="company_link"/>
    <xs:enumeration value="on_site"/>
    <xs:enumeration value="online"/>
  </xs:restriction>
</xs:simpleType>
</xs:schema>
```

payment_registered

Een seintje van CRM (bijv. via het kassasysteem) dat een factuur handmatig is voldaan, zodat wij deze in FossBilling op 'Betaald' kunnen zetten.

XML

```xml
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>a23bc45d-89ef-1234-b567-1f03c3d4e580</message_id>
    <version>2.0</version>
    <type>payment_registered</type>
    <timestamp>2026-03-29T18:35:00Z</timestamp>
    <source>crm_system</source>
    <correlation_id>f47ac10b-58cc-4372-a567-0e02b2c3d479</correlation_id>
  </header>
<body>
  <invoice>
    <id>INV-2026-001</id>
    <status>paid</status>
    <amount_paid currency="eur">0.00</amount_paid>
    <due_date>2026-03-29</due_date>
  </invoice>
  <transaction>
    <id>CUST-12345</id>
    <payment_method>online</payment_method>
  </transaction>
</body>
</message>
```

XSD

```xml
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" elementFormDefault="qualified">

  <!-- payment_registered XSD -->

  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
```

```
      <xs:element name="body" type="PaymentRegisteredBodyType"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>

<xs:complexType name="HeaderType">
  <xs:sequence>
    <xs:element name="message_id" type="UuidV4"/>
    <xs:element name="version" type="xs:string"/>
    <xs:element name="type" type="xs:string" fixed="payment_registered"/>
    <xs:element name="timestamp" type="xs:dateTime"/>
    <xs:element name="source" type="xs:string"/>
    <xs:element name="correlation_id" type="UuidV4"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="PaymentRegisteredBodyType">
  <xs:sequence>
    <xs:element name="invoice" type="InvoiceType"/>
    <xs:element name="transaction" type="TransactionType"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="InvoiceType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="status" type="InvoiceStatusType"/>
    <xs:element name="amount_paid" type="AmountType"/>
    <xs:element name="due_date" type="xs:date"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="TransactionType">
  <xs:sequence>
```

```
    <xs:element name="id" type="xs:string"/>
    <xs:element name="payment_method" type="PaymentMethodType"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="AmountType">
  <xs:simpleContent>
    <xs:extension base="xs:decimal">
      <xs:attribute name="currency" type="CurrencyCodeType" use="required"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>

<xs:simpleType name="UuidV4">
  <xs:restriction base="xs:string">
    <xs:pattern value="[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}"/>
  </xs:restriction>
</xs:simpleType>

<xs:simpleType name="InvoiceStatusType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="paid"/>
    <xs:enumeration value="pending"/>
    <xs:enumeration value="cancelled"/>
  </xs:restriction>
</xs:simpleType>

<xs:simpleType name="PaymentMethodType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="company_link"/>
    <xs:enumeration value="on_site"/>
    <xs:enumeration value="online"/>
  </xs:restriction>
</xs:simpleType>
```

```
  <xs:simpleType name="CurrencyCodeType">
    <xs:restriction base="xs:string">
      <xs:pattern value="[a-z]{3}"/>
    </xs:restriction>
  </xs:simpleType>

</xs:schema>
```

new_registration

Wordt direct bij een nieuwe inschrijving gestuurd om de klant aan te maken en de eenmalige registratiekost te factureren.

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>550e8400-e29b-41d4-a716-446655440000</message_id>
    <version>2.0</version>
    <type>new_registration</type>
    <timestamp>2026-04-06T14:30:00Z</timestamp>
    <source>crm_system</source>
  </header>
  <body>
    <customer>
      <id>CUST-9999</id>
      <email>jan.janssens@example.com</email>
      <is_company_linked>false</is_company_linked>
      <company_name></company_name>
      <address>
        <street>Kerkstraat</street>
        <number>10</number>
        <postal_code>1000</postal_code>
        <city>Brussel</city>
        <country>België</country>
      </address>
```

```
    </customer>
    <registration_fee currency="eur">25.00</registration_fee>
  </body>
</message>
```

XSD

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
elementFormDefault="qualified">

  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="message_id" type="xs:string"/>
              <xs:element name="version" type="xs:string" fixed="2.0"/>
              <xs:element name="type" type="xs:string" fixed="new_registration"/>
              <xs:element name="timestamp" type="xs:dateTime"/>
              <xs:element name="source" type="xs:string"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>

        <xs:element name="body">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="customer">
                <xs:complexType>
                  <xs:sequence>
                    <xs:element name="id" type="xs:string"/>
                    <xs:element name="email" type="xs:string"/>
                    <xs:element name="is_company_linked" type="xs:boolean"/>
                    <xs:element name="company_name" type="xs:string" minOccurs="0"/>
```

```
<xs:element name="address">
<xs:complexType>
<xs:sequence>
<xs:element name="street" type="xs:string"/>
<xs:element name="number" type="xs:string"/>
<xs:element name="postal_code" type="xs:string"/>
<xs:element name="city" type="xs:string"/>
<xs:element name="country" type="xs:string"/>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="registration_fee">
<xs:complexType>
<xs:simpleContent>
<xs:extension base="xs:decimal">
<xs:attribute name="currency" type="xs:string" use="required"/>
</xs:extension>
</xs:simpleContent>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>

</xs:schema>
```

Van Facturatie (Uitgaand)

Facturatie => mailing

invoice_created_notification

XML

```xml
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>uuid</message_id>
    <version>1.0</version>
    <type>invoice_created_notification</type>
    <timestamp>2026-04-01T18:00:00Z</timestamp>
    <source>facturatie_system</source>
  </header>
  <body>
    <recipient_email>klant@example.com</recipient_email>
    <invoice_id>INV-2026-001</invoice_id>
    <subject>Uw nieuwe factuur</subject>
    <message_text>Beste klant, uw factuur is beschikbaar.</message_text>
    <pdf_url>https://fossbilling/invoices/INV-2026-001.pdf</pdf_url>
  </body>
</message>
```

XSD

```xml
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
          elementFormDefault="qualified">
  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
        <xs:element name="body" type="MailingNotificationBodyType"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
  <xs:complexType name="HeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="xs:string"/>
      <xs:element name="version" type="xs:string"/>
```

```
    <xs:element name="type" type="xs:string" fixed="invoice_created_notification"/>
    <xs:element name="timestamp" type="xs:dateTime"/>
    <xs:element name="source" type="xs:string"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="MailingNotificationBodyType">
  <xs:sequence>
    <xs:element name="recipient_email" type="xs:string"/>
    <xs:element name="invoice_id" type="xs:string"/>
    <xs:element name="subject" type="xs:string"/>
    <xs:element name="message_text" type="xs:string"/>
    <xs:element name="pdf_url" type="xs:anyURI"/>
  </xs:sequence>
</xs:complexType>
</xs:schema>
```

Facturatie => CRM

send_invoice

Stuurt de gegenereerde factuurdata terug naar CRM, inclusief de PDF-link, factuurnummer en vervaldatum voor jullie database.

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>c91df23a-47be-6789-d012-3e25f5a6b702</message_id>
    <version>2.0</version>
    <type>send_invoice</type>
    <timestamp>2026-03-29T18:36:00Z</timestamp>
    <source>facturatie_system_01</source>
    <correlation_id>f47ac10b-58cc-4372-a567-0e02b2c3d479</correlation_id>
  </header>
  <body>
    <customer>
      <id>12345</id>
```

```
      <email>info@bedrijf.be</email>
      <is_company_linked>true</is_company_linked>
      <company_id>FOSS-CUST-102</company_id>
      <address>
        <street>Kiekenmarkt</street>
        <number>42</number>
        <postal_code>1000</postal_code>
        <city>Brussel</city>
        <country>be</country>
      </address>
    </customer>
    <invoice>
      <id>INV-2026-001</id>
      <status>paid</status>
      <amount_paid currency="eur">15.00</amount_paid>
      <due_date>2026-03-06</due_date>
      <pdf_url>https://fossbilling.intern/invoices/INV-2026-001.pdf</pdf_url>
    </invoice>
  </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
elementFormDefault="qualified">


  <!-- send_invoice XSD -->


  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
        <xs:element name="body" type="SendInvoiceBodyType"/>
      </xs:sequence>
    </xs:complexType>
```

```
</xs:element>

<xs:complexType name="HeaderType">
  <xs:sequence>
    <xs:element name="message_id" type="UuidV4"/>
    <xs:element name="version" type="xs:string"/>
    <xs:element name="type" type="xs:string" fixed="send_invoice"/>
    <xs:element name="timestamp" type="xs:dateTime"/>
    <xs:element name="source" type="xs:string"/>
    <xs:element name="correlation_id" type="UuidV4"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="SendInvoiceBodyType">
  <xs:sequence>
    <xs:element name="customer" type="CustomerType"/>
    <xs:element name="invoice" type="InvoiceType"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="CustomerType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="email" type="xs:string"/>
    <xs:element name="is_company_linked" type="xs:boolean"/>
    <xs:element name="company_id" type="xs:string" minOccurs="0"/>
    <xs:element name="address" type="AddressType"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="AddressType">
  <xs:sequence>
    <xs:element name="street" type="xs:string"/>
    <xs:element name="number" type="xs:string"/>
```

```
    <xs:element name="postal_code" type="xs:string"/>
    <xs:element name="city" type="xs:string"/>
    <xs:element name="country" type="CountryCodeType"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="InvoiceType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="status" type="xs:string" fixed="paid"/>
    <xs:element name="amount_paid" type="AmountType"/>
    <xs:element name="due_date" type="xs:date"/>
    <xs:element name="pdf_url" type="xs:anyURI"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="AmountType">
  <xs:simpleContent>
    <xs:extension base="xs:decimal">
      <xs:attribute name="currency" type="CurrencyCodeType" use="required"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>

<xs:simpleType name="UuidV4">
  <xs:restriction base="xs:string">
    <xs:pattern value="[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}"/>
  </xs:restriction>
</xs:simpleType>

<xs:simpleType name="CountryCodeType">
  <xs:restriction base="xs:string">
    <xs:pattern value="[a-z]{2}"/>
  </xs:restriction>
```

```
  </xs:simpleType>

  <xs:simpleType name="CurrencyCodeType">
    <xs:restriction base="xs:string">
      <xs:pattern value="[a-z]{3}"/>
    </xs:restriction>
  </xs:simpleType>

</xs:schema>
```

invoice_cancelled

Een notificatie dat een factuur is geannuleerd (en gecrediteerd), zodat de status in CRM bijgewerkt kan worden.

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>e12bc34d-56fa-7890-b123-4c56d7e8f901</message_id>
    <version>2.0</version>
    <type>invoice_cancelled</type>
    <timestamp>2026-03-29T18:40:00Z</timestamp>
    <source>facturatie_system_01</source>
    <correlation_id>a23bc45d-89ef-1234-b567-1f03c3d4e580</correlation_id>
  </header>

  <body>
    <customer>
      <id>12345</id>
      <email>info@bedrijf.be</email>
      <is_company_linked>true</is_company_linked>
      <company_id>FOSS-CUST-102</company_id>
      <address>
        <street>Kiekenmarkt</street>
        <number>42</number>
        <postal_code>1000</postal_code>
```

```
      <city>Brussel</city>
      <country>be</country>
    </address>
  </customer>

  <invoice>
    <id>INV-2026-001</id>
    <status>cancelled</status>
    <amount_paid currency="eur">0.00</amount_paid>
    <due_date>2026-03-06</due_date>
    <cancellation_reason>customer_request</cancellation_reason>
  </invoice>
 </body>
</message>
```

XSD

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
elementFormDefault="qualified">

 <!-- Root -->
 <xs:element name="message">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="header" type="HeaderType"/>
      <xs:element name="body" type="InvoiceCancelledBodyType"/>
    </xs:sequence>
  </xs:complexType>
 </xs:element>

 <!-- Header -->
 <xs:complexType name="HeaderType">
  <xs:sequence>
    <xs:element name="message_id" type="UuidV4"/>
    <xs:element name="version" type="xs:string"/>
```

```
    <xs:element name="type" type="xs:string" fixed="invoice_cancelled"/>
    <xs:element name="timestamp" type="xs:dateTime"/>
    <xs:element name="source" type="xs:string"/>
    <xs:element name="correlation_id" type="UuidV4"/>
  </xs:sequence>
</xs:complexType>

<!-- Body -->
<xs:complexType name="InvoiceCancelledBodyType">
  <xs:sequence>
    <xs:element name="customer" type="CustomerType"/>
    <xs:element name="invoice" type="InvoiceType"/>
  </xs:sequence>
</xs:complexType>

<!-- Customer -->
<xs:complexType name="CustomerType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="email" type="xs:string"/>
    <xs:element name="is_company_linked" type="xs:boolean"/>
    <xs:element name="company_id" type="xs:string" minOccurs="0"/>
    <xs:element name="address" type="AddressType"/>
  </xs:sequence>
</xs:complexType>

<!-- Address -->
<xs:complexType name="AddressType">
  <xs:sequence>
    <xs:element name="street" type="xs:string"/>
    <xs:element name="number" type="xs:string"/>
    <xs:element name="postal_code" type="xs:string"/>
    <xs:element name="city" type="xs:string"/>
    <xs:element name="country" type="CountryCodeType"/>
```

```
  </xs:sequence>
</xs:complexType>

<!-- Invoice -->
<xs:complexType name="InvoiceType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="status" type="InvoiceStatusType"/>
    <xs:element name="amount_paid" type="AmountType"/>
    <xs:element name="due_date" type="xs:date"/>
    <xs:element name="cancellation_reason" type="xs:string"/>
  </xs:sequence>
</xs:complexType>

<!-- Shared simple types -->
<xs:complexType name="AmountType">
  <xs:simpleContent>
    <xs:extension base="xs:decimal">
      <xs:attribute name="currency" type="CurrencyCodeType" use="required"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>

<xs:simpleType name="UuidV4">
  <xs:restriction base="xs:string">
    <xs:pattern value="[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}"/>
  </xs:restriction>
</xs:simpleType>

<xs:simpleType name="InvoiceStatusType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="cancelled"/>
    <xs:enumeration value="paid"/>
    <xs:enumeration value="pending"/>
```

```
    </xs:restriction>
  </xs:simpleType>

  <xs:simpleType name="CountryCodeType">
    <xs:restriction base="xs:string">
      <xs:pattern value="[a-z]{2}"/>
    </xs:restriction>
  </xs:simpleType>

  <xs:simpleType name="CurrencyCodeType">
    <xs:restriction base="xs:string">
      <xs:pattern value="[a-z]{3}"/>
    </xs:restriction>
  </xs:simpleType>

</xs:schema>
```

payment_registered

Bevestiging vanuit FossBilling dat een betaling succesvol is verwerkt.

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>a23bc45d-89ef-1234-b567-1f03c3d4e580</message_id>
    <version>2.0</version>
    <type>payment_registered</type>
    <timestamp>2026-03-29T18:35:00Z</timestamp>
    <source>crm_system</source>
    <correlation_id>f47ac10b-58cc-4372-a567-0e02b2c3d479</correlation_id>
  </header>
<body>
  <invoice>
    <id>INV-2026-001</id>
    <status>paid</status>
    <amount_paid currency="eur">0.00</amount_paid>
```

```
    <due_date>2026-03-29</due_date>
  </invoice>
  <transaction>
    <id>CUST-12345</id>
    <payment_method>online</payment_method>
  </transaction>
</body>
</message>
```

XSD

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" elementFormDefault="qualified">

  <!-- payment_registered XSD -->

  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
        <xs:element name="body" type="PaymentRegisteredBodyType"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>

  <xs:complexType name="HeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="UuidV4"/>
      <xs:element name="version" type="xs:string"/>
      <xs:element name="type" type="xs:string" fixed="payment_registered"/>
      <xs:element name="timestamp" type="xs:dateTime"/>
      <xs:element name="source" type="xs:string"/>
      <xs:element name="correlation_id" type="UuidV4"/>
    </xs:sequence>
```

```
</xs:complexType>

<xs:complexType name="PaymentRegisteredBodyType">
  <xs:sequence>
    <xs:element name="invoice" type="InvoiceType"/>
    <xs:element name="transaction" type="TransactionType"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="InvoiceType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="status" type="InvoiceStatusType"/>
    <xs:element name="amount_paid" type="AmountType"/>
    <xs:element name="due_date" type="xs:date"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="TransactionType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="payment_method" type="PaymentMethodType"/>
  </xs:sequence>
</xs:complexType>

<xs:complexType name="AmountType">
  <xs:simpleContent>
    <xs:extension base="xs:decimal">
      <xs:attribute name="currency" type="CurrencyCodeType" use="required"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>

<xs:simpleType name="UuidV4">
```

```
    <xs:restriction base="xs:string">
      <xs:pattern value="[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}"/>
    </xs:restriction>
  </xs:simpleType>

  <xs:simpleType name="InvoiceStatusType">
    <xs:restriction base="xs:string">
      <xs:enumeration value="paid"/>
      <xs:enumeration value="pending"/>
      <xs:enumeration value="cancelled"/>
    </xs:restriction>
  </xs:simpleType>

  <xs:simpleType name="PaymentMethodType">
    <xs:restriction base="xs:string">
      <xs:enumeration value="company_link"/>
      <xs:enumeration value="on_site"/>
      <xs:enumeration value="online"/>
    </xs:restriction>
  </xs:simpleType>

  <xs:simpleType name="CurrencyCodeType">
    <xs:restriction base="xs:string">
      <xs:pattern value="[a-z]{3}"/>
    </xs:restriction>
  </xs:simpleType>

</xs:schema>
```

Kassa

Naar Kassa (Inkomend)

CRM → Kassa (Queue: kassa.incoming)

new_registration

XML

```xml
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>MSG-CRM-1001</message_id>
    <type>new_registration</type>
    <source>crm</source>
    <timestamp>2026-02-24T16:00:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <customer>
      <email>info@techbedrijf.be</email>
      <contact>
        <first_name>Jan</first_name>
        <last_name>Peeters</last_name>
      </contact>
      <company_name>TechCompany NV</company_name>
      <type>company</type>
      <vat_number>BE0123456789</vat_number>
      <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
      <date_of_birth>1995-06-15</date_of_birth>
    </customer>
    <payment_due>
      <amount>50.00</amount>
      <status>unpaid</status>
    </payment_due>
  </body>
</message>
```

XSD

```xml
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="HeaderType">
    <xs:sequence>
```

```
    <xs:element name="message_id" type="xs:string"/>
    <xs:element name="type" type="xs:string" fixed="new_registration"/>
    <xs:element name="source" type="xs:string"/>
    <xs:element name="timestamp" type="xs:dateTime"/>
    <xs:element name="version" type="xs:string"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="CustomerType">
  <xs:sequence>
    <xs:element name="email" type="xs:string"/>
    <xs:complexType name="ContactType">
      <xs:sequence>
        <xs:element name="first_name" type="xs:string"/>
        <xs:element name="last_name" type="xs:string"/>
      </xs:sequence>
    </xs:complexType>
    <xs:element name="contact" type="ContactType"/>
    <xs:element name="company_name" type="xs:string" minOccurs="0"/>
    <xs:element name="type">
      <xs:simpleType>
        <xs:restriction base="xs:string">
          <xs:enumeration value="private"/>
          <xs:enumeration value="company"/>
        </xs:restriction>
      </xs:simpleType>
    </xs:element>
    <xs:element name="vat_number" type="xs:string" minOccurs="0"/>
    <xs:element name="user_id" type="xs:string"/>
    <xs:element name="date_of_birth" type="xs:date"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PaymentDueType">
  <xs:sequence>
    <xs:element name="amount" type="xs:decimal"/>
```

```
      <xs:element name="status">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:enumeration value="unpaid"/>
            <xs:enumeration value="paid"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
        <xs:element name="body">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="customer" type="CustomerType"/>
              <xs:element name="payment_due" type="PaymentDueType"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

profile_update

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>MSG-CRM-7788</message_id>
    <type>profile_update</type>
    <source>crm</source>
```

```
    <timestamp>2026-02-24T19:15:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
    <email>info@techbedrijf.be</email>
    <contact>
      <first_name>Jan</first_name>
      <last_name>Peeters</last_name>
    </contact>
    <company_name>TechCompany NV</company_name>
    <date_of_birth>1995-06-15</date_of_birth>
    <type>company</type>
    <vat_number>BE0123456789</vat_number>
  </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="HeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="xs:string"/>
      <xs:element name="type" type="xs:string" fixed="profile_update"/>
      <xs:element name="source" type="xs:string"/>
      <xs:element name="timestamp" type="xs:dateTime"/>
      <xs:element name="version" type="xs:string"/>
    </xs:sequence>
  </xs:complexType>
  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
        <xs:element name="body">
          <xs:complexType>
```

```
            <xs:sequence>
              <xs:element name="user_id" type="xs:string"/>
              <xs:element name="email" type="xs:string"/>
              <xs:complexType name="ContactType">
                <xs:sequence>
                  <xs:element name="first_name" type="xs:string"/>
                  <xs:element name="last_name" type="xs:string"/>
                </xs:sequence>
              </xs:complexType>
              <xs:element name="contact" type="ContactType"/>
              <xs:element name="type">
                <xs:simpleType>
                  <xs:restriction base="xs:string">
                    <xs:enumeration value="private"/>
                    <xs:enumeration value="company"/>
                  </xs:restriction>
                </xs:simpleType>
              </xs:element>
              <xs:element name="company_name" type="xs:string" minOccurs="0"/>
              <xs:element name="vat_number" type="xs:string" minOccurs="0"/>
              <xs:element name="date_of_birth" type="xs:date"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```


cancel_registration

XML


```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>f12e3d4c-5b6a-7d8e-9f0a-1b2c3d4e5f6a</message_id>
```

```
    <type>cancel_registration</type>
    <source>crm</source>
    <timestamp>2026-03-04T11:05:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
    <session_id>session-uuid-001</session_id>
  </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="HeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="xs:string"/>
      <xs:element name="type" type="xs:string" fixed="cancel_registration"/>
      <xs:element name="source" type="xs:string"/>
      <xs:element name="timestamp" type="xs:dateTime"/>
      <xs:element name="version" type="xs:string"/>
    </xs:sequence>
  </xs:complexType>
  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
        <xs:element name="body">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="user_id" type="xs:string"/>
              <xs:element name="session_id" type="xs:string"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
```

```
    </xs:sequence>
   </xs:complexType>
  </xs:element>
</xs:schema>
```

IoT → Kassa (Queue: kassa.incoming)

badge_scanned

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>MSG-IOT-5544</message_id>
    <type>badge_scanned</type>
    <source>iot_scanner_bar</source>
    <timestamp>2026-02-24T19:15:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <badge_id>QR-98765-ABC</badge_id>
    <location>hoofdbar</location>
  </body>
</message>
```

XSD

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="HeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="xs:string"/>
      <xs:element name="type" type="xs:string" fixed="badge_scanned"/>
      <xs:element name="source" type="xs:string"/>
      <xs:element name="timestamp" type="xs:dateTime"/>
      <xs:element name="version" type="xs:string"/>
    </xs:sequence>
  </xs:complexType>
  <xs:element name="message">
```

```
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
        <xs:element name="body">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="badge_id" type="xs:string"/>
              <xs:element name="location" type="xs:string"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

Van Kassa (Uitgaand)

Kassa → CRM (Exchange: kassa.exchange)

consumption_order (Routing Key: kassa.payments.consumption)

XML (Aankoop gekoppeld aan user)

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>f47ac10b-58cc-4372-a567-0e02b2c3d479</message_id>
    <type>consumption_order</type>
    <source>kassa</source>
    <timestamp>2026-02-24T18:30:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <is_anonymous>false</is_anonymous>
    <customer>
      <id>12345</id>
      <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
      <type>company</type>
```

```
      <email>info@bedrijf.be</email>
      <address>
        <street>Kiekenmarkt</street>
        <number>42</number>
        <postal_code>1000</postal_code>
        <city>Brussel</city>
        <country>be</country>
      </address>
    </customer>
    <items>
      <item>
        <id>LINE-4201</id>
        <sku>BEV-001</sku>
        <description>Koffie</description>
        <quantity>2</quantity>
        <unit_price currency="eur">2.50</unit_price>
        <total_amount currency="eur">5.00</total_amount>
        <vat_rate>6</vat_rate>
      </item>
    </items>
  </body>
</message>
```


XML (Anonieme aankoop)


```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>f11a0000-0000-0000-0000-000000000001</message_id>
    <type>consumption_order</type>
    <source>kassa</source>
    <timestamp>2026-04-15T15:00:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <is_anonymous>true</is_anonymous>
```

```
    <items>
      <item>
        <id>LINE-4202</id>
        <sku>BEV-002</sku>
        <description>Cola</description>
        <quantity>1</quantity>
        <unit_price currency="eur">2.00</unit_price>
        <total_amount currency="eur">2.00</total_amount>
        <vat_rate>6</vat_rate>
      </item>
    </items>
  </body>
</message>
```


XML (Top-up, user zet geld op badge)


```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>f13c0000-0000-0000-0000-000000000099</message_id>
    <type>consumption_order</type>
    <source>kassa</source>
    <timestamp>2026-04-15T11:30:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <is_anonymous>false</is_anonymous>
    <customer>
      <id>12345</id>
      <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
      <type>private</type>
      <email>jan@peeters.be</email>
    </customer>
    <items>
      <item>
        <id>LINE-4210</id>
```

```
      <sku>TOPUP-010</sku>
      <description>Top-up EUR 10</description>
      <quantity>1</quantity>
      <unit_price currency="eur">10.00</unit_price>
      <total_amount currency="eur">10.00</total_amount>
      <vat_rate>0</vat_rate>
      <item_type>wallet_topup</item_type>
    </item>
  </items>
 </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
 <xs:complexType name="HeaderType">
  <xs:sequence>
   <xs:element name="message_id" type="xs:string"/>
   <xs:element name="type" type="xs:string" fixed="consumption_order"/>
   <xs:element name="source" type="xs:string"/>
   <xs:element name="timestamp" type="xs:dateTime"/>
   <xs:element name="version" type="xs:string"/>
  </xs:sequence>
 </xs:complexType>
 <xs:complexType name="AddressType">
  <xs:sequence>
   <xs:element name="street" type="xs:string"/>
   <xs:element name="number" type="xs:string"/>
   <xs:element name="postal_code" type="xs:string"/>
   <xs:element name="city" type="xs:string"/>
   <xs:element name="country" type="xs:string"/>
  </xs:sequence>
 </xs:complexType>
 <xs:complexType name="CustomerType">
  <xs:sequence>
```

```
    <xs:element name="id" type="xs:string" minOccurs="0"/>
    <xs:element name="user_id" type="xs:string" minOccurs="0"/>
    <xs:element name="type">
      <xs:simpleType>
        <xs:restriction base="xs:string">
          <xs:enumeration value="private"/>
          <xs:enumeration value="company"/>
        </xs:restriction>
      </xs:simpleType>
    </xs:element>
    <xs:element name="email" type="xs:string" minOccurs="0"/>
    <xs:element name="address" type="AddressType" minOccurs="0"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ItemType">
  <xs:sequence>
    <xs:element name="id" type="xs:string"/>
    <xs:element name="sku" type="xs:string"/>
    <xs:element name="description" type="xs:string"/>
    <xs:element name="quantity" type="xs:positiveInteger"/>
    <xs:element name="unit_price">
      <xs:complexType>
        <xs:simpleContent>
          <xs:extension base="xs:decimal">
            <xs:attribute name="currency" type="xs:string" use="required"/>
          </xs:extension>
        </xs:simpleContent>
      </xs:complexType>
    </xs:element>
    <xs:element name="vat_rate">
      <xs:simpleType>
        <xs:restriction base="xs:integer">
          <xs:enumeration value="0"/>
          <xs:enumeration value="6"/>
```

```
          <xs:enumeration value="12"/>
          <xs:enumeration value="21"/>
        </xs:restriction>
      </xs:simpleType>
    </xs:element>
    <xs:element name="total_amount">
      <xs:complexType>
        <xs:simpleContent>
          <xs:extension base="xs:decimal">
            <xs:attribute name="currency" type="xs:string" use="required"/>
          </xs:extension>
        </xs:simpleContent>
      </xs:complexType>
    </xs:element>
    <xs:element name="item_type" type="xs:string" minOccurs="0"/>
  </xs:sequence>
</xs:complexType>
<xs:element name="message">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="header" type="HeaderType"/>
      <xs:element name="body">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="is_anonymous" type="xs:boolean" minOccurs="0"
default="false"/>
            <xs:element name="customer" type="CustomerType" minOccurs="0"/>
            <xs:element name="items">
              <xs:complexType>
                <xs:sequence>
                  <xs:element name="item" type="ItemType" maxOccurs="unbounded"/>
                </xs:sequence>
              </xs:complexType>
            </xs:element>
          </xs:sequence>
```

```
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
 </xs:element>
</xs:schema>
```

payment_registered (Verschillende routing keys per context)

XML (Consumptie) (Routing Key: kassa.payments.consumption)

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
 <header>
  <message_id>a23bc45d-89ef-1234-b567-1f03c3d4e580</message_id>
  <type>payment_registered</type>
  <source>kassa</source>
  <timestamp>2026-02-24T18:35:00Z</timestamp>
  <version>2.0</version>
  <correlation_id>f47ac10b-58cc-4372-a567-0e02b2c3d479</correlation_id>
 </header>
 <body>
  <payment_context>consumption</payment_context>
  <invoice>
   <id>INV-2026-001</id>
   <status>paid</status>
   <amount_paid currency="eur">15.00</amount_paid>
   <due_date>2026-02-24</due_date>
  </invoice>
  <transaction>
   <id>TRX-987654</id>
   <payment_method>company_link</payment_method>
  </transaction>
 </body>
</message>
```

XML (Registratie) (Routing Key: kassa.payments.registration)

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<message>
  <header>
    <message_id>f14d0000-0000-0000-0000-000000000004</message_id>
    <type>payment_registered</type>
    <source>kassa</source>
    <timestamp>2026-04-15T09:15:00Z</timestamp>
    <version>2.0</version>
    <correlation_id>MSG-CRM-1001</correlation_id>
  </header>
  <body>
    <payment_context>registration</payment_context>
    <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
    <invoice>
      <status>paid</status>
      <amount_paid currency="eur">50.00</amount_paid>
      <due_date>2026-04-15</due_date>
    </invoice>
    <transaction>
      <id>TRX-2026-04150001</id>
      <payment_method>on_site</payment_method>
    </transaction>
  </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="HeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="xs:string"/>
      <xs:element name="type" type="xs:string" fixed="payment_registered"/>
      <xs:element name="source" type="xs:string"/>
      <xs:element name="timestamp" type="xs:dateTime"/>
      <xs:element name="version" type="xs:string"/>
      <xs:element name="correlation_id" type="xs:string" minOccurs="0"/>
```

```
  </xs:sequence>
</xs:complexType>
<xs:complexType name="CurrencyAmountType">
  <xs:simpleContent>
    <xs:extension base="xs:decimal">
      <xs:attribute name="currency" type="xs:string" use="required"/>
    </xs:extension>
  </xs:simpleContent>
</xs:complexType>
<xs:element name="message">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="header" type="HeaderType"/>
      <xs:element name="body">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="payment_context">
              <xs:simpleType>
                <xs:restriction base="xs:string">
                  <xs:enumeration value="registration"/>
                  <xs:enumeration value="consumption"/>
                </xs:restriction>
              </xs:simpleType>
            </xs:element>
            <xs:element name="user_id" type="xs:string" minOccurs="0"/>
            <xs:element name="invoice">
              <xs:complexType>
                <xs:sequence>
                  <xs:element name="id" type="xs:string" minOccurs="0"/>
                  <xs:element name="status">
                    <xs:simpleType>
                      <xs:restriction base="xs:string">
                        <xs:enumeration value="paid"/>
                        <xs:enumeration value="pending"/>
```

```
                <xs:enumeration value="cancelled"/>
              </xs:restriction>
            </xs:simpleType>
          </xs:element>
          <xs:element name="amount_paid" type="CurrencyAmountType"/>
          <xs:element name="due_date" type="xs:date"/>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
    <xs:element name="transaction">
      <xs:complexType>
        <xs:sequence>
          <xs:element name="id" type="xs:string"/>
          <xs:element name="payment_method">
            <xs:simpleType>
              <xs:restriction base="xs:string">
                <xs:enumeration value="company_link"/>
                <xs:enumeration value="on_site"/>
                <xs:enumeration value="online"/>
              </xs:restriction>
            </xs:simpleType>
          </xs:element>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:schema>
```

invoice_request (Routing Key: kassa.payments.invoice)

XML

```xml
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>b12c3d4e-5f6a-7890-bcde-f01234567890</message_id>
    <type>invoice_request</type>
    <source>kassa</source>
    <timestamp>2026-02-24T20:00:00Z</timestamp>
    <version>2.0</version>
    <correlation_id>f47ac10b-58cc-4372-a567-0e02b2c3d479</correlation_id>
  </header>
  <body>
    <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
    <invoice_data>
      <first_name>Jan</first_name>
      <last_name>Peeters</last_name>
      <email>jan@peeters.be</email>
      <address>
        <street>Kiekenmarkt</street>
        <number>42</number>
        <postal_code>1000</postal_code>
        <city>Brussel</city>
        <country>be</country>
      </address>
      <vat_number>BE0123456789</vat_number>
    </invoice_data>
  </body>
</message>
```


XSD


```xml
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="HeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="xs:string"/>
      <xs:element name="type" type="xs:string" fixed="invoice_request"/>
```

```
    <xs:element name="source" type="xs:string"/>
    <xs:element name="timestamp" type="xs:dateTime"/>
    <xs:element name="version" type="xs:string"/>
    <xs:element name="correlation_id" type="xs:string" minOccurs="0"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="AddressType">
  <xs:sequence>
    <xs:element name="street" type="xs:string"/>
    <xs:element name="number" type="xs:string"/>
    <xs:element name="postal_code" type="xs:string"/>
    <xs:element name="city" type="xs:string"/>
    <xs:element name="country" type="xs:string"/>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="InvoiceDataType">
  <xs:sequence>
    <xs:element name="first_name" type="xs:string"/>
    <xs:element name="last_name" type="xs:string"/>
    <xs:element name="email" type="xs:string"/>
    <xs:element name="address" type="AddressType"/>
    <xs:element name="vat_number" type="xs:string" minOccurs="0"/>
  </xs:sequence>
</xs:complexType>
<xs:element name="message">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="header" type="HeaderType"/>
      <xs:element name="body">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="user_id" type="xs:string"/>
            <xs:element name="invoice_data" type="InvoiceDataType"/>
          </xs:sequence>
```

```
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
 </xs:element>
</xs:schema>
```

badge_assigned (Routing Key: kassa.payments.badge)

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>f12b0000-0000-0000-0000-000000000002</message_id>
    <type>badge_assigned</type>
    <source>kassa</source>
    <timestamp>2026-04-15T09:05:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <badge_id>BADGE-RF-00142</badge_id>
    <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
    <assigned_at>2026-04-15T09:05:00Z</assigned_at>
  </body>
</message>
```

XSD

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="                                        ">
  <xs:complexType name="HeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="xs:string"/>
      <xs:element name="type" type="xs:string" fixed="badge_assigned"/>
      <xs:element name="source" type="xs:string"/>
      <xs:element name="timestamp" type="xs:dateTime"/>
      <xs:element name="version" type="xs:string"/>
    </xs:sequence>
```

```
  </xs:complexType>
  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
        <xs:element name="body">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="badge_id" type="xs:string"/>
              <xs:element name="user_id" type="xs:string"/>
              <xs:element name="assigned_at" type="xs:dateTime"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

refund_processed (Routing Key: kassa.payments.refund)

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>f15e0000-0000-0000-0000-000000000005</message_id>
    <type>refund_processed</type>
    <source>kassa</source>
    <timestamp>2026-04-15T16:45:00Z</timestamp>
    <version>2.0</version>
    <correlation_id>f14d0000-0000-0000-0000-000000000004</correlation_id>
  </header>
  <body>
    <refund_type>consumption_item</refund_type>
    <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
    <refund>
```

```
    <amount currency="eur">5.00</amount>
    <method>badge_wallet</method>
    <reason>duplicate_payment</reason>
    <description>Dubbele aanrekening gecorrigeerd door kassamedewerker</description>
  </refund>
  <original_transaction_id>TRX-2026-04150001</original_transaction_id>
  <new_wallet_balance currency="eur">20.50</new_wallet_balance>
 </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="                                                    ">
 <xs:complexType name="HeaderType">
  <xs:sequence>
    <xs:element name="message_id" type="xs:string"/>
    <xs:element name="type" type="xs:string" fixed="refund_processed"/>
    <xs:element name="source" type="xs:string"/>
    <xs:element name="timestamp" type="xs:dateTime"/>
    <xs:element name="version" type="xs:string"/>
    <xs:element name="correlation_id" type="xs:string" minOccurs="0"/>
  </xs:sequence>
 </xs:complexType>
 <xs:complexType name="CurrencyAmountType">
  <xs:simpleContent>
    <xs:extension base="xs:decimal">
      <xs:attribute name="currency" type="xs:string" use="required"/>
    </xs:extension>
  </xs:simpleContent>
 </xs:complexType>
 <xs:element name="message">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="header" type="HeaderType"/>
```

```
<xs:element name="body">
<xs:complexType>
<xs:sequence>
<xs:element name="refund_type">
<xs:simpleType>
<xs:restriction base="xs:string">
<xs:enumeration value="consumption_item"/>
<xs:enumeration value="partial"/>
</xs:restriction>
</xs:simpleType>
</xs:element>
<xs:element name="user_id" type="xs:string" minOccurs="0"/>
<xs:element name="refund">
<xs:complexType>
<xs:sequence>
<xs:element name="amount" type="CurrencyAmountType"/>
<xs:element name="method">
<xs:simpleType>
<xs:restriction base="xs:string">
<xs:enumeration value="badge_wallet"/>
<xs:enumeration value="cash"/>
<xs:enumeration value="card_reversal"/>
</xs:restriction>
</xs:simpleType>
</xs:element>
<xs:element name="reason">
<xs:simpleType>
<xs:restriction base="xs:string">
<xs:enumeration value="duplicate_payment"/>
<xs:enumeration value="customer_request"/>
<xs:enumeration value="system_error"/>
</xs:restriction>
</xs:simpleType>
</xs:element>
```

```
                <xs:element name="description" type="xs:string" minOccurs="0"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
        <xs:element name="original_transaction_id" type="xs:string"/>
        <xs:element name="new_wallet_balance" type="CurrencyAmountType" minOccurs="0"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:schema>
```

Kassa → Monitoring (Exchange: kassa.exchange)

system_error (Routing Key: kassa.errors)

XML (Algemene error)

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>c9d2e415-5f6a-4b7c-8e1d-2a3b4c5d6e7f</message_id>
    <type>system_error</type>
    <source>kassa</source>
    <timestamp>2026-02-24T19:25:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <error_code>invalid_xml_format</error_code>
    <error_description>Message does not comply with schema_new_registration.xsd</error_description>
    <related_message_id>MSG-CRM-1001</related_message_id>
  </body>
</message>
```

XML (Badge niet gevonden)

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<message>
  <header>
    <message_id>f16f0000-0000-0000-0000-000000000006</message_id>
    <type>system_error</type>
    <source>kassa</source>
    <timestamp>2026-04-15T14:55:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <error_code>badge_not_found</error_code>
    <error_description>Badge BADGE-RF-99999 niet gevonden in lokale Odoo-cache.
</error_description>
    <related_message_id>MSG-IOT-5523</related_message_id>
  </body>
</message>
```


XSD

Kassa → Frontend (Exchange: kassa.exchange)

payment_status (Routing Key: kassa.frontend.payment)

XML


```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>d98a7c65-4b5e-4c6f-8d9e-1a2b3c4d5e6f</message_id>
    <type>payment_status</type>
    <source>kassa</source>
    <timestamp>2026-03-04T10:15:30Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
    <payment_status>paid</payment_status>
  </body>
</message>
```


XSD

```xml
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="                                    ">
  <xs:complexType name="HeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="xs:string"/>
      <xs:element name="type" type="xs:string" fixed="payment_status"/>
      <xs:element name="source" type="xs:string"/>
      <xs:element name="timestamp" type="xs:dateTime"/>
      <xs:element name="version" type="xs:string"/>
    </xs:sequence>
  </xs:complexType>
  <xs:element name="message">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="header" type="HeaderType"/>
        <xs:element name="body">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="user_id" type="xs:string"/>
              <xs:element name="payment_status">
                <xs:simpleType>
                  <xs:restriction base="xs:string">
                    <xs:enumeration value="paid"/>
                    <xs:enumeration value="pending"/>
                  </xs:restriction>
                </xs:simpleType>
              </xs:element>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

wallet_balance_update (Routing Key: kassa.frontend.wallet)

XML (Reguliere update)

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>e54a8b72-1c2d-3e4f-5678-7a8b9c0d1e2f</message_id>
    <type>wallet_balance_update</type>
    <source>kassa</source>
    <timestamp>2026-03-06T20:30:00Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
    <wallet_balance>15.50</wallet_balance>
  </body>
</message>
```

XML (Top-up, user zet geld op badge)

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>f13d0000-0000-0000-0000-000000000100</message_id>
    <type>wallet_balance_update</type>
    <source>kassa</source>
    <timestamp>2026-04-15T11:30:05Z</timestamp>
    <version>2.0</version>
  </header>
  <body>
    <user_id>e8b27c1d-4f2a-4b3e-9c5f-123456789abc</user_id>
    <wallet_balance>25.50</wallet_balance>
  </body>
</message>
```

XSD

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
```

```
<xs:complexType name="HeaderType">
  <xs:sequence>
    <xs:element name="message_id" type="xs:string"/>
    <xs:element name="type" type="xs:string" fixed="wallet_balance_update"/>
    <xs:element name="source" type="xs:string"/>
    <xs:element name="timestamp" type="xs:dateTime"/>
    <xs:element name="version" type="xs:string"/>
  </xs:sequence>
</xs:complexType>
<xs:element name="message">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="header" type="HeaderType"/>
      <xs:element name="body">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="user_id" type="xs:string"/>
            <xs:element name="wallet_balance" type="xs:decimal"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
</xs:schema>
```

Frontend

Architechtuur

Het frontend-systeem (Drupal) communiceert via RabbitMQ met de andere subsystemen. Het verstuurt XML-berichten naar het CRM, Planning en Monitoring, en ontvangt berichten van Planning, Kassa en CRM.

Verstuurd door Frontend (Senders)

Frontend → CRM, Planning, Mailing

user.unregistered

Gebruiker schrijft zich uit voor een sessie. Fan-out naar: crm.salesforce, planning.outlook, mailing.sendgrid Sender: UserUnregisteredSender.phpegistratie in Salesforce. Queue: crm.incoming Sender: NewRegistrationSender.php

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>{uuid}</message_id>
    <timestamp>2026-04-04T10:00:00+02:00</timestamp>
    <source>frontend.drupal</source>
    <receiver>crm.salesforce planning.outlook mailing.sendgrid</receiver>
    <type>user.unregistered</type>
    <version>1.0</version>
    <correlation_id></correlation_id>
  </header>
  <body>
    <user_id>user-001</user_id>
    <session_id>sess-42</session_id>
    <timestamp>2026-04-04T10:00:00+02:00</timestamp>
  </body>
</message>
```

XSD

Verplicht: user_id en session_id. Het bericht wordt identiek naar alle drie queues gestuurd via fan-out exchange.

Frontend → Monitoring

user.checkin

Badge check-in event. Queue: user.checkin Sender: UserCheckinSender.php (v1.0)

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>{uuid}</message_id>
    <timestamp>2026-04-04T09:30:00+02:00</timestamp>
    <source>frontend.drupal</source>
    <receiver>monitoring.elastic</receiver>
```

```
    <type>user.checkin</type>
    <version>1.0</version>
    <correlation_id></correlation_id>
  </header>
  <body>
    <user_id>user-001</user_id>
    <badge_id>BADGE-XYZ-001</badge_id>
    <timestamp>2026-04-04T09:30:00+02:00</timestamp>
  </body>
</message>
```

XSD

Verplicht: user_id en badge_id. Verstuurd naar monitoring.elastic voor real-time dashboarding.

Frontend → Planning

calender.invite

Kalenderuitnodiging naar Planning. Exchange: calendar.exchange (topic, durable). Routing key: calendar.invite Sender: CalendarInviteSender.php

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>{uuid}</message_id>
    <timestamp>2026-04-04T10:00:00+00:00</timestamp>
    <source>frontend</source>
    <type>calendar.invite</type>
  </header>
  <body>
    <session_id>sess-42</session_id>
    <title>Keynote: AI 2026</title>
    <start_datetime>2026-04-04T09:00:00Z</start_datetime>
    <end_datetime>2026-04-04T10:00:00Z</end_datetime>
    <location>Zaal A</location> <!-- optioneel -->
  </body>
</message>
```

XSD

Verplicht: session_id, title, start_datetime, end_datetime. location is optioneel. Gebruikt topic exchange met namespace "urn:integration:planning:v1". Planning consumer luistert op queue planning.calendar.invite. Let op: deze header is vereenvoudigd ten opzichte van de standaard v1.0-header — version, receiver en correlation_id zijn afwezig conform de Planning-XSD.

Frontend → CRM

new_registration

Registratie in Salesforce. Queue: crm.incoming Sender: NewRegistrationSender.php

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
  <header>
    <message_id>{UUID v4}</message_id>
    <version>2.0</version>
    <type>new_registration</type>
    <timestamp>{ISO 8601 UTC}</timestamp>
    <source>frontend</source>
    <correlation_id>{UUID v4}</correlation_id> <!-- optioneel, als laatste -->
  </header>
  <body>...</body>
</message>
```

v1.0 Header (overige senders) — met xmlns="urn:integration:planning:v1":

```
<?xml version="1.0" encoding="UTF-8"?>
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>{UUID v4}</message_id>
    <timestamp>{ISO 8601}</timestamp>
    <source>frontend.drupal</source>
    <receiver>{doelsysteem}</receiver>
    <type>{berichttype}</type>
    <version>1.0</version>
    <correlation_id></correlation_id>
  </header>
```

```
  <body>...</body>
</message>
```

XSD

user_id, email, first_name, last_name zijn verplicht.

type moet "private" of "company" zijn.

Bij company zijn company_name en vat_number vereist.

Geen xmlns attribuut op <message>.

Header volgorde verplicht: message_id → version → type → timestamp → source.

user.created

Nieuw gebruikersaccount aangemaakt. Queue: user.created Sender: UserCreatedSender.php

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>{uuid}</message_id>
    <timestamp>2026-04-04T10:00:00+02:00</timestamp>
    <source>frontend.drupal</source>
    <receiver>crm.salesforce</receiver>
    <type>user.created</type>
    <version>1.0</version>
    <correlation_id></correlation_id>
  </header>
  <body>
    <user>
      <first_name>Jan</first_name>
      <last_name>Peeters</last_name>
      <email>jan@example.com</email>
      <is_company>false</is_company>
    </user>
  </body>
</message>
```

XSD

Verplicht: email, first_name, last_name.

user.registered

Gebruiker schrijft zich in voor een sessie. Queue: frontend.user.registered

Sender: UserRegisteredSender.php

XML

```xml
<?xml version="1.0" encoding="UTF-8"?>
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>{uuid}</message_id>
    <timestamp>2026-04-04T10:00:00+02:00</timestamp>
    <source>frontend.drupal</source>
    <receiver>crm.salesforce</receiver>
    <type>user.registered</type>
    <version>1.0</version>
    <correlation_id></correlation_id>
  </header>
  <body>
    <user>
      <first_name>Jan</first_name>
      <last_name>Peeters</last_name>
      <email>jan@example.com</email>
      <is_company>false</is_company>
    </user>
    <session>
      <session_id>sess-42</session_id>
      <name>Keynote: AI 2026</name>
    </session>
    <payment_status>pending</payment_status>
  </body>
</message>
```

XSD

Verplicht: email, first_name, last_name, session.session_id, session.name. Bij company is vat_number verplicht. payment_status is altijd "pending" bij initieel versturen.

Ontvangen door Frontend (Receivers)

Planning → Frontend

session_created

Nieuwe sessie aangemaakt door Planning. Exchange: planning.exchange (topic, durable). Routing key: planning.session.created. Queue: frontend.planning.session.created (durable) Receiver: SessionCreatedReceiver.php

XML

```
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>msg-uuid</message_id>
    <timestamp>2026-04-04T09:00:00Z</timestamp>
    <source>planning</source>
    <type>session_created</type>
    <version>1.0</version> <!-- optioneel -->
    <correlation_id>corr-uuid-here</correlation_id> <!-- optioneel -->
  </header>
  <body>
    <session_id>sess-42</session_id>
    <title>Keynote: AI 2026</title>
    <start_datetime>2026-04-04T09:00:00Z</start_datetime>
    <end_datetime>2026-04-04T10:00:00Z</end_datetime>
    <location>Zaal A</location> <!-- optioneel -->
    <session_type>keynote</session_type> <!-- optioneel -->
    <status>planned</status> <!-- optioneel -->
    <max_attendees>200</max_attendees> <!-- optioneel, default 0 -->
    <current_attendees>0</current_attendees> <!-- optioneel, default 0 -->
  </body>
</message>
```

XSD

Verplicht: session_id, title, start_datetime, end_datetime. Type-waarde is session_created (underscore, conform Planning). Ondersteunt zowel namespaced als non-namespaced XML. Integer velden: max_attendees, current_attendees (default: 0).

session_updated

Sessie-update ontvangen. Queue: frontend.session.update (durable) Receiver: SessionUpdateReceiver.php

XML

```
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>msg-uuid</message_id>
    <timestamp>2026-04-04T09:30:00Z</timestamp>
    <source>planning</source>
    <type>session_updated</type>
    <version>1.0</version> <!-- optioneel -->
    <correlation_id>corr-uuid-here</correlation_id> <!-- optioneel -->
  </header>
  <body>
    <session_id>sess-42</session_id>
    <title>Keynote: AI 2026 (Updated)</title> <!-- optioneel -->
    <start_datetime>2026-04-04T11:00:00Z</start_datetime> <!-- optioneel -->
    <end_datetime>2026-04-04T12:00:00Z</end_datetime> <!-- optioneel -->
    <location>Zaal B</location> <!-- optioneel -->
    <session_type>keynote</session_type> <!-- optioneel -->
    <status>published</status> <!-- optioneel -->
    <max_attendees>150</max_attendees> <!-- optioneel -->
    <current_attendees>25</current_attendees> <!-- optioneel -->
  </body>
</message>
```

XSD

Verplicht: session_id. Type-waarde is session_updated (underscore, conform Planning). Alle overige body-velden zijn optioneel. De receiver verwerkt de volledige Planning-body inclusief title, start_datetime, end_datetime, session_type, status, max_attendees en current_attendees.

IoT/Kassa → Frontend

badge_scanned

Badge scan event ontvangen. Queue: badge.scanned (durable) Receiver: BadgeScannedReceiver.php

XML

```
<message>
  <header>
    ...
    <type>badge_scanned</type>
    ...
  </header>
  <body>
    <badge_id>BADGE-XYZ-001</badge_id>
    <location>hoofdbar</location> <!-- optioneel -->
  </body>
</message>
```

XSD

Verplicht: badge_id. user_id

Kassa → Frontend

payment_registered

Betalingsbevestiging ontvangen. Queue: payment.registered (durable) Receiver: PaymentRegisteredReceiver.php

XML

```
<message>
  <header>
    ...
    <type>payment_registered</type>
    ...
  </header>
  <body>
    <payment_context>registration</payment_context> <!-- "registration" of "consumption" -->
    <user_id>user-001</user_id> <!-- optioneel -->
    <invoice>
      <id>INV-2026-001</id> <!-- optioneel -->
      <status>paid</status>
      <amount_paid currency="eur">50.00</amount_paid>
```

```
    <due_date>2026-04-15</due_date>
  </invoice>
  <transaction>
    <id>TRX-987654</id>
    <payment_method>on_site</payment_method>
  </transaction>
 </body>
</message>
```

XSD

Verplicht: invoice.status, transaction.payment_method

CRM→ Frontend

wordt nog geverifieerd

...

XML

XSD

Algemene Berichtstructuur

v2.0 Header (NewRegistrationSender) — zonder xmlns attribuut:

```
<?xml version="1.0" encoding="UTF-8"?>
<message>
 <header>
  <message_id>{UUID v4}</message_id>
  <version>2.0</version>
  <type>new_registration</type>
  <timestamp>{ISO 8601 UTC}</timestamp>
  <source>frontend</source>
  <correlation_id>{UUID v4}</correlation_id> <!-- optioneel, als laatste -->
 </header>
```

```
  <body>...</body>
</message>
```

v1.0 Header (overige senders) — met xmlns="urn:integration:planning:v1":

```
<?xml version="1.0" encoding="UTF-8"?>
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>{UUID v4}</message_id>
    <timestamp>{ISO 8601}</timestamp>
    <source>frontend.drupal</source>
    <receiver>{doelsysteem}</receiver>
    <type>{berichttype}</type>
    <version>1.0</version>
    <correlation_id></correlation_id>
  </header>
  <body>...</body>
</message>
```

Infra

Monitoring

heartbeat

Alle teams > monitoring

XML

```
<heartbeat>
  <system>facturatie</system>
  <timestamp>2026-03-15T10:30:01Z</timestamp>
  <uptime>1500</uptime>
</heartbeat>
```

Planning

calendar.invite (incoming)

XML

```
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>msg-uuid</message_id>
    <timestamp>2026-05-15T09:00:00Z</timestamp>
    <source>frontend</source>
    <type>calendar.invite</type>
  </header>
  <body>
    <session_id>sess-uuid-001</session_id>
    <title>Keynote: AI in Healthcare</title>
    <start_datetime>2026-05-15T14:00:00Z</start_datetime>
    <end_datetime>2026-05-15T15:00:00Z</end_datetime>
    <location>online</location>
  </body>
</message>
```

XSD

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    targetNamespace="urn:integration:planning:v1"
    xmlns="urn:integration:planning:v1"
    elementFormDefault="qualified"
    attributeFormDefault="unqualified">

    <xs:element name="message" type="CalendarInviteMessageType"/>

    <xs:complexType name="CalendarInviteMessageType">
        <xs:sequence>
            <xs:element name="header" type="CalendarInviteHeaderType"/>
            <xs:element name="body" type="CalendarInviteBodyType"/>
        </xs:sequence>
    </xs:complexType>

    <xs:complexType name="CalendarInviteHeaderType">
```

```
      <xs:sequence>
        <xs:element name="message_id" type="xs:string"/>
        <xs:element name="timestamp" type="xs:dateTime"/>
        <xs:element name="source" type="xs:string"/>
        <xs:element name="type">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:enumeration value="calendar.invite"/>
            </xs:restriction>
          </xs:simpleType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>

    <xs:complexType name="CalendarInviteBodyType">
      <xs:sequence>
        <xs:element name="session_id" type="xs:string"/>
        <xs:element name="title" type="xs:string"/>
        <xs:element name="start_datetime" type="xs:dateTime"/>
        <xs:element name="end_datetime" type="xs:dateTime"/>
        <xs:element name="location" type="xs:string" minOccurs="0"/>
      </xs:sequence>
    </xs:complexType>

  </xs:schema>
```

session_created

XML

```
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>550e8400-e29b-41d4-a716-446655440000</message_id>
    <timestamp>2026-05-15T09:00:00Z</timestamp>
    <source>planning</source>
    <type>session_created</type>
    <version>1.0</version>
```

```
    <correlation_id>corr-uuid-here</correlation_id>
  </header>
  <body>
    <session_id>sess-uuid-001</session_id>
    <title>Keynote: AI in Healthcare</title>
    <start_datetime>2026-05-15T14:00:00Z</start_datetime>
    <end_datetime>2026-05-15T15:00:00Z</end_datetime>
    <location>Aula A - Campus Jette</location>
    <session_type>keynote</session_type>
    <status>published</status>
    <max_attendees>120</max_attendees>
    <current_attendees>0</current_attendees>
  </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    targetNamespace="urn:integration:planning:v1"
    xmlns="urn:integration:planning:v1"
    elementFormDefault="qualified"
    attributeFormDefault="unqualified">

    <xs:element name="message" type="SessionCreatedMessageType"/>

    <xs:complexType name="SessionCreatedMessageType">
        <xs:sequence>
            <xs:element name="header" type="SessionCreatedHeaderType"/>
            <xs:element name="body" type="SessionCreatedBodyType"/>
        </xs:sequence>
    </xs:complexType>

    <xs:complexType name="SessionCreatedHeaderType">
        <xs:sequence>
```

```
            <xs:element name="message_id" type="xs:string"/>
            <xs:element name="timestamp" type="xs:dateTime"/>
            <xs:element name="source" type="xs:string"/>
            <xs:element name="type">
              <xs:simpleType>
                <xs:restriction base="xs:string">
                  <xs:enumeration value="session_created"/>
                </xs:restriction>
              </xs:simpleType>
            </xs:element>
            <xs:element name="version" type="xs:string" minOccurs="0"/>
            <xs:element name="correlation_id" type="xs:string" minOccurs="0"/>
          </xs:sequence>
        </xs:complexType>

        <xs:complexType name="SessionCreatedBodyType">
          <xs:sequence>
            <xs:element name="session_id" type="xs:string"/>
            <xs:element name="title" type="xs:string"/>
            <xs:element name="start_datetime" type="xs:string"/>
            <xs:element name="end_datetime" type="xs:string"/>
            <xs:element name="location" type="xs:string" minOccurs="0"/>
            <xs:element name="session_type" type="xs:string" minOccurs="0"/>
            <xs:element name="status" type="xs:string" minOccurs="0"/>
            <xs:element name="max_attendees" type="xs:nonNegativeInteger" minOccurs="0"/>
            <xs:element name="current_attendees" type="xs:nonNegativeInteger" minOccurs="0"/>
          </xs:sequence>
        </xs:complexType>

      </xs:schema>
```

session_updated

XML

```
<message xmlns="urn:integration:planning:v1">
  <header>
```

```
    <message_id>550e8400-e29b-41d4-a716-446655440000</message_id>
    <timestamp>2026-05-15T09:30:00Z</timestamp>
    <source>planning</source>
    <type>session_updated</type>
    <version>1.0</version>
    <correlation_id>corr-uuid-here</correlation_id>
  </header>
  <body>
    <session_id>sess-uuid-001</session_id>
    <title>Keynote: AI in Healthcare (Updated)</title>
    <start_datetime>2026-05-15T14:30:00Z</start_datetime>
    <end_datetime>2026-05-15T15:30:00Z</end_datetime>
    <location>Aula A - Campus Jette</location>
    <session_type>keynote</session_type>
    <status>published</status>
    <max_attendees>150</max_attendees>
    <current_attendees>25</current_attendees>
  </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    targetNamespace="urn:integration:planning:v1"
    xmlns="urn:integration:planning:v1"
    elementFormDefault="qualified"
    attributeFormDefault="unqualified">

    <xs:element name="message" type="SessionUpdatedMessageType"/>

    <xs:complexType name="SessionUpdatedMessageType">
        <xs:sequence>
            <xs:element name="header" type="SessionUpdatedHeaderType"/>
            <xs:element name="body" type="SessionUpdatedBodyType"/>
```

```
    </xs:sequence>
  </xs:complexType>

  <xs:complexType name="SessionUpdatedHeaderType">
    <xs:sequence>
      <xs:element name="message_id" type="xs:string"/>
      <xs:element name="timestamp" type="xs:dateTime"/>
      <xs:element name="source" type="xs:string"/>
      <xs:element name="type">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:enumeration value="session_updated"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:element>
      <xs:element name="version" type="xs:string" minOccurs="0"/>
      <xs:element name="correlation_id" type="xs:string" minOccurs="0"/>
    </xs:sequence>
  </xs:complexType>

  <xs:complexType name="SessionUpdatedBodyType">
    <xs:sequence>
      <xs:element name="session_id" type="xs:string"/>
      <xs:element name="title" type="xs:string"/>
      <xs:element name="start_datetime" type="xs:string"/>
      <xs:element name="end_datetime" type="xs:string"/>
      <xs:element name="location" type="xs:string" minOccurs="0"/>
      <xs:element name="session_type" type="xs:string" minOccurs="0"/>
      <xs:element name="status" type="xs:string" minOccurs="0"/>
      <xs:element name="max_attendees" type="xs:nonNegativeInteger" minOccurs="0"/>
      <xs:element name="current_attendees" type="xs:nonNegativeInteger" minOccurs="0"/>
    </xs:sequence>
  </xs:complexType>
```

```
</xs:schema>
```

session_deleted

XML

```
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>550e8400-e29b-41d4-a716-446655440000</message_id>
    <timestamp>2026-05-15T10:00:00Z</timestamp>
    <source>planning</source>
    <type>session_deleted</type>
    <version>1.0</version>
    <correlation_id>corr-uuid-here</correlation_id>
  </header>
  <body>
    <session_id>sess-uuid-001</session_id>
    <reason>cancelled</reason>
    <deleted_by>planning-admin</deleted_by>
  </body>
</message>
```

XSD

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    targetNamespace="urn:integration:planning:v1"
    xmlns="urn:integration:planning:v1"
    elementFormDefault="qualified"
    attributeFormDefault="unqualified">

    <xs:element name="message" type="SessionDeletedMessageType"/>

    <xs:complexType name="SessionDeletedMessageType">
        <xs:sequence>
            <xs:element name="header" type="SessionDeletedHeaderType"/>
            <xs:element name="body" type="SessionDeletedBodyType"/>
        </xs:sequence>
```

```
    </xs:complexType>

    <xs:complexType name="SessionDeletedHeaderType">
      <xs:sequence>
        <xs:element name="message_id" type="xs:string"/>
        <xs:element name="timestamp" type="xs:dateTime"/>
        <xs:element name="source" type="xs:string"/>
        <xs:element name="type">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:enumeration value="session_deleted"/>
            </xs:restriction>
          </xs:simpleType>
        </xs:element>
        <xs:element name="version" type="xs:string" minOccurs="0"/>
        <xs:element name="correlation_id" type="xs:string" minOccurs="0"/>
      </xs:sequence>
    </xs:complexType>

    <xs:complexType name="SessionDeletedBodyType">
      <xs:sequence>
        <xs:element name="session_id" type="xs:string"/>
        <xs:element name="reason" type="xs:string" minOccurs="0"/>
        <xs:element name="deleted_by" type="xs:string" minOccurs="0"/>
      </xs:sequence>
    </xs:complexType>

  </xs:schema>
```

session_view_request

XML

```
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>550e8400-e29b-41d4-a716-446655440000</message_id>
    <timestamp>2026-05-15T10:05:00Z</timestamp>
```

```
    <source>planning</source>
    <type>session_view_request</type>
    <version>1.0</version>
    <correlation_id>corr-uuid-here</correlation_id>
  </header>
  <body>
    <session_id>sess-uuid-001</session_id>
  </body>
</message>
```

XSD

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
    xmlns:xs="http://www.w3.org/2001/XMLSchema"
    targetNamespace="urn:integration:planning:v1"
    xmlns="urn:integration:planning:v1"
    elementFormDefault="qualified"
    attributeFormDefault="unqualified">

    <xs:element name="message" type="SessionViewRequestMessageType"/>

    <xs:complexType name="SessionViewRequestMessageType">
        <xs:sequence>
            <xs:element name="header" type="SessionViewRequestHeaderType"/>
            <xs:element name="body" type="SessionViewRequestBodyType"/>
        </xs:sequence>
    </xs:complexType>

    <xs:complexType name="SessionViewRequestHeaderType">
        <xs:sequence>
            <xs:element name="message_id" type="xs:string"/>
            <xs:element name="timestamp" type="xs:dateTime"/>
            <xs:element name="source" type="xs:string"/>
            <xs:element name="type">
                <xs:simpleType>
```

```
                <xs:restriction base="xs:string">
                  <xs:enumeration value="session_view_request"/>
                </xs:restriction>
              </xs:simpleType>
            </xs:element>
            <xs:element name="version" type="xs:string" minOccurs="0"/>
            <xs:element name="correlation_id" type="xs:string" minOccurs="0"/>
          </xs:sequence>
        </xs:complexType>

        <xs:complexType name="SessionViewRequestBodyType">
          <xs:sequence>
            <xs:element name="session_id" type="xs:string" minOccurs="0"/>
          </xs:sequence>
        </xs:complexType>

      </xs:schema>
```

session_view_response

XML

```
<message xmlns="urn:integration:planning:v1">
  <header>
    <message_id>550e8400-e29b-41d4-a716-446655440000</message_id>
    <timestamp>2026-05-15T10:05:01Z</timestamp>
    <source>planning</source>
    <type>session_view_response</type>
    <version>1.0</version>
    <correlation_id>corr-uuid-here</correlation_id>
  </header>
  <body>
    <request_message_id>req-uuid-001</request_message_id>
    <requested_session_id>sess-uuid-001</requested_session_id>
    <status>ok</status>
    <session_count>1</session_count>
    <sessions>
```

```
    <session>
      <session_id>sess-uuid-001</session_id>
      <title>Keynote: AI in Healthcare</title>
      <start_datetime>2026-05-15T14:00:00Z</start_datetime>
      <end_datetime>2026-05-15T15:00:00Z</end_datetime>
      <location>Aula A - Campus Jette</location>
      <session_type>keynote</session_type>
      <status>published</status>
      <max_attendees>120</max_attendees>
      <current_attendees>25</current_attendees>
    </session>
  </sessions>
 </body>
</message>
```


XSD


```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  targetNamespace="urn:integration:planning:v1"
  xmlns="urn:integration:planning:v1"
  elementFormDefault="qualified"
  attributeFormDefault="unqualified">

  <xs:element name="message" type="SessionViewResponseMessageType"/>

  <xs:complexType name="SessionViewResponseMessageType">
    <xs:sequence>
      <xs:element name="header" type="SessionViewResponseHeaderType"/>
      <xs:element name="body" type="SessionViewResponseBodyType"/>
    </xs:sequence>
  </xs:complexType>

  <xs:complexType name="SessionViewResponseHeaderType">
    <xs:sequence>
```

```
        <xs:element name="message_id" type="xs:string"/>
        <xs:element name="timestamp" type="xs:dateTime"/>
        <xs:element name="source" type="xs:string"/>
        <xs:element name="type">
            <xs:simpleType>
                <xs:restriction base="xs:string">
                    <xs:enumeration value="session_view_response"/>
                </xs:restriction>
            </xs:simpleType>
        </xs:element>
        <xs:element name="version" type="xs:string" minOccurs="0"/>
        <xs:element name="correlation_id" type="xs:string" minOccurs="0"/>
    </xs:sequence>
</xs:complexType>

<xs:complexType name="SessionViewResponseBodyType">
    <xs:sequence>
        <xs:element name="request_message_id" type="xs:string"/>
        <xs:element name="requested_session_id" type="xs:string" minOccurs="0"/>
        <xs:element name="status">
            <xs:simpleType>
                <xs:restriction base="xs:string">
                    <xs:enumeration value="ok"/>
                    <xs:enumeration value="not_found"/>
                </xs:restriction>
            </xs:simpleType>
        </xs:element>
        <xs:element name="session_count" type="xs:nonNegativeInteger"/>
        <xs:element name="sessions" type="SessionsType"/>
    </xs:sequence>
</xs:complexType>

<xs:complexType name="SessionsType">
    <xs:sequence>
```

```
        <xs:element name="session" type="SessionType" minOccurs="0"
maxOccurs="unbounded"/>
      </xs:sequence>
    </xs:complexType>

    <xs:complexType name="SessionType">
      <xs:sequence>
        <xs:element name="session_id" type="xs:string"/>
        <xs:element name="title" type="xs:string" minOccurs="0"/>
        <xs:element name="start_datetime" type="xs:string" minOccurs="0"/>
        <xs:element name="end_datetime" type="xs:string" minOccurs="0"/>
        <xs:element name="location" type="xs:string" minOccurs="0"/>
        <xs:element name="session_type" type="xs:string" minOccurs="0"/>
        <xs:element name="status" type="xs:string" minOccurs="0"/>
        <xs:element name="max_attendees" type="xs:nonNegativeInteger" minOccurs="0"/>
        <xs:element name="current_attendees" type="xs:nonNegativeInteger" minOccurs="0"/>
      </xs:sequence>
    </xs:complexType>

</xs:schema>
```